

Gauge Group Origin from F4 Dynkin Diagram Removal

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We extend our topological F4 analysis to map out the detailed mechanism of electroweak and strong gauge group generation. By applying Dynkin diagram node removal techniques to the F4 algebra, we show that the $SU(3) \times SU(2) \times U(1)$ symmetry group is the unique stable residual gauge symmetry compatible with the 24-cell lattice deformation.

I. INTRODUCTION

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\mathfrak{f}_4 \xrightarrow{-\alpha_4} \mathfrak{b}_4 \xrightarrow{-\alpha_1} \mathfrak{a}_3 \times \mathfrak{u}_1 \cong \mathfrak{su}(3) \times \mathfrak{su}(2) \times \mathfrak{u}(1) \quad (1)$$

III. ALGEBRAIC REDUCTIONS

Removing specific roots corresponds to projection steps that align with the spatial six-cone geometry, establishing the three gauge couplings without free parameters.

IV. CONCLUSION

The origin of gauge groups in the Standard Model is traditionally treated as an experimental input. In this work, we derive this structure from the F4 Dynkin diagram.

The Standard Model is the only mathematically allowed low-energy gauge group under the topological constraints of the 24-cell lattice.

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